Lecture 28.01: Introduction to Decision Trees

📌 Topic Overview

In this lecture, we understand what a **Decision Tree** is, why it is one of the most intuitive and widely used machine learning models, and how it makes predictions using a sequence of simple if-else decisions.

💡 Real-World Intuition: The Loan Approval Analogy

Imagine a bank loan officer evaluating whether an applicant should be granted a loan. The officer naturally follows a chain of questions:

1. *Is the applicant's loan amount* ≥ 1 *Lakh?*
2. *If yes, is the applicant employed?*
3. *If employed, approve the loan; otherwise, reject.*

In Python code, this thinking looks exactly like simple conditional statements:

```
if loan_amt >= 100000:
if employed == "Yes":
print("Loan Approved")
else:
print("Loan Rejected")
else:
print("Loan Approved")
```

A **Decision Tree** is simply this nested `if-else` decision logic learned automatically from data!

🌳 Tree Structural Anatomy

When we draw this sequence of decisions as a flowchart, it forms an upside-down tree:

```
flowchart TD
A["Root Node: Loan Amount >= 1L?"] -->|"Yes"| B["Internal Node: Employed == Yes?"]
A -->|"No"| C["Leaf Node: Approved (Yes)"]
B -->|"Yes"| D["Leaf Node: Approved (Yes)"]
B -->|"No"| E["Leaf Node: Rejected (No)"]
```

1. Root Node

- The very first node at the top where the decision process starts.
- It contains all the training samples before any split occurs.

2. Internal (Decision) Nodes

- Intermediate nodes where a feature is tested (e.g., `employed == 'Yes'?`).
- Each internal node splits the data further down two or more branches.

3. Branches (Edges)

- The connecting arrows representing the possible answers to the condition (e.g., `Yes` vs `No`).

4. Leaf (Terminal) Nodes

- The final endpoints where no further questions are asked.
- Each leaf provides the final predicted output (e.g., class label in classification or average value in regression).

🎯 Visualizing Data Partitioning: Blue Dots vs Orange Dots

Consider a mixed dataset with two classes:

- **Blue dots** = Positive class (e.g., Loan Approved = Yes)
- **Orange dots** = Negative class (e.g., Loan Approved = No)

At the root node, blue and orange dots are mixed together (**impure**). When the tree tests a feature (for example, `Gender` → Male vs Female):

- One branch receives a group that has mostly orange dots.
- The other branch receives a group that has mostly blue dots.

By continuing this splitting recursively, the tree isolates the samples into **pure leaf nodes** containing only blue or only orange dots.

🏛️ Famous Decision Tree Algorithms

Over the years, several algorithms have been developed to build decision trees:

1. **CART (Classification and Regression Trees)**: The standard algorithm used in Scikit-Learn. It produces binary splits (2 children per node).
2. **ID3 (Iterative Dichotomiser 3)**: Developed by Ross Quinlan; uses Information Gain (Entropy).
3. **C4.5**: Quinlan's successor to ID3 that supports continuous features and pruning.
4. **CHAID (Chi-square Automatic Interaction Detector)**: Uses Chi-square tests for categorical multi-way splits.

💻 Python Implementation

```
import numpy as np
from sklearn.tree import DecisionTreeClassifier

# Feature matrix: [Loan_Amount_in_Lakhs, Employed (1=Yes, 0=No)]
X = np.array([
[0.5, 0],
[0.8, 1],
[1.5, 1],
[2.0, 0],
[3.0, 1]
])
# Target: 1 = Approved, 0 = Rejected
y = np.array([1, 1, 1, 0, 1])

# Initialize and fit Decision Tree Classifier
clf = DecisionTreeClassifier(random_state=42)
clf.fit(X, y)

# Predict for a new applicant: Loan 2.5 Lakh, Unemployed (0)
prediction = clf.predict([[2.5, 0]])
print("Prediction for [2.5L, Unemployed]:", "Approved" if prediction[0] == 1 else
```

🔑 Key Takeaways

1. Decision Trees predict outcomes by asking a sequence of hierarchical if-then questions.
2. The tree consists of a **Root Node**, **Internal Decision Nodes**, **Branches**, and **Leaf Nodes**.
3. The goal of tree building is to partition mixed data points into clean, homogeneous (pure) groups.
4. Scikit-Learn implements the **CART** algorithm by default.